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POKHARA COLLEGE OF MANAGEMENT
(Affiliated to Pokhara University)
Nadipur-3, Pokhara



A Major Project Report
On
“GoInventory”

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August, 2023

BONAFIDE CERTIFICATE

Certified that this project report

Entitled:

“A REPORT ON GOINVENTORY”

Is the bonafide work of

Prashanti Bhattachan

in partial fulfilment of the requirements for the Bachelor in Computer and Information System (BCIS) embodies the bonafide work done by them under my supervision.

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ACKNOWLEDGEMENT

I'd want to use this opportunity to express my heartfelt gratitude for everybody who has helped with the Go Inventory in any way.

To begin, I must express my gratitude to Er. Hari Prasad Baral, Er. Rupesh Kumar Shah, Er. Raj Kiran Chatkhuli and Pokhara College of Management for being a constant source of inspiration and assistance in assisting me prepare the project, as well as for their insight, expertise, encouragement, and guidance in outlining views on studying the project's specifics and acquiring the proper project vision.

I would also want to express my appreciation and gratitude to the researchers, specialists, and writers, as well as their works, which proved to be extremely beneficial to the project.

Finally, I want to thank all of my friends for their never-ending encouragement, support, and love.

The project would not have been accomplished without the contributions of the persons named above.

ABSTRACT

In today's digital era, the fast expansion of organizations demands effective and simplified inventory management solutions. The goal of this project is to design and construct an easy-to-use Inventory Management Website to assist organizations in successfully tracking, managing, and optimizing their inventory procedures. The website will have a user-friendly design that will allow users to quickly input, update, and retrieve inventory information.

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LIST OF ABBREVIATIONS

Pokhara University = PU

Pokhara College of Management = PCM

Content Management System = CMS

Hyper Text Markup Language = HTML

Cascading Style Sheets = CSS

Unified Modeling Language = UML

Entity Relationship Diagram = ERD

Data Flow Diagram = DFD

FIFO =First-In, First-Out

LIFO = Last-In, First-Out

CHAPTER-I

INTRODUCTION

1.1 Background

Inventory management is the act of monitoring and regulating the movement, storage, and accessibility of products or materials inside a company or organization. To make sure that the correct items are available at the right time, in the right quantities, and at the right price, it entails a variety of tasks like ordering, storing, tracking, and maintaining inventory levels. For organizations to satisfy consumer needs, cut expenses, and improve operations, effective inventory management is essential. It attempts to achieve a balance between preventing stockouts (items running out) and reducing excess inventory (overstocking), which can tie up capital and result in additional expenditures.

Important elements of inventory control include:

- Demand forecasting: It is the process of estimating future product demand based on historical data, market trends, and other pertinent variables to determine inventory levels.
- Inventory Control: Implementing procedures and systems to monitor and maintain ideal stock levels, including establishing lead times, safety stock levels, and reorder points.
- Stock replenishment: Starting production or purchase orders to restock inventories based on anticipated demand and existing stock levels.
- Stock tracking: It involves tracking inventory goods using techniques like barcoding, RFID (Radio Frequency Identification), or other tracking technologies.
- Inventory valuation: It is the process of giving the stock you have on hand a monetary value for financial reporting reasons. This may be done using techniques like FIFO (First-In, First-Out) or LIFO (Last-In, First-Out).

1.2 Problem of Statements:

Stockouts, excess inventory, erroneous forecasts, and human mistakes are problems caused by the ineffectiveness of our company's present inventory management method, which also lacks real-time insight. As an online business, we face the following specific problems:

- There is Inventory visibility in real time is lacking.
- It becomes inadequate order fulfillment.
- Sometimes demand forecasting errors may occurs.
- Procurement and manual reordering.
- It becomes improper platform integration for e-commerce.

1.3 Objectives:

The Objective of our website is to provide:

- To efficiently organize and manage ordered products, staffs details, and product information.

1.4 Scope

The scope of online inventory management in an immense undertaking involves setting up an entire system to track, regulate, and optimize inventories in an online setting. It involves setting up real-time awareness of inventory levels and locations, speeding order fulfillment processes, predicting consumer demand, and optimizing inventory levels. The initiative intends to coordinate inventory across several sales channels, save transportation expenses, prevent stockouts, and boost operational effectiveness. In order to make educated decisions, it also entails producing reliable reports and statistics. The project aims to increase client satisfaction, enhance profitability, and ensure smooth inventory operations throughout the project long-term through the use of technology along with efficient inventory management techniques.

CHAPTER-II

LITERATURE REVIEW

2.1 Existing system vs Purpose system

2.1.1 Existing system

Bidhee Pvt. Ltd. was founded with the goal of developing ideas into digital platforms by meeting clients' specific needs. We are now one of Nepal's top firms for creating and delivering innovative mobile and web technology products and services. Its main goal is to digitalize customer demands, build software to help clients achieve their goals, and produce an innovative solution to help organizations improve their day-to-day performance and decision making.

Bidhee is a market leader in the implementation of ERP Solutions for the Construction Industry, Sales ERP Solutions for the Manufacturing and Trading Industry, and HR Payroll Solutions in hundreds of Nepalese organizations. Bidhee also offers Website Design, eCommerce Solutions, and Mobile Application Development for the Android and iOS platforms.

2.1.2 Purposed System:

There is currently an online service called Bidhee that performs a similar role as our system. However, unlike Bidhee, our website "GoInventory" offers several advantages and features that aid in faster and easier inventory managements.

CHAPTER-III

SYSTEM REQUIREMENTS

3.1 Software Requirements

The tools used in making this system are as follows:

- HTML, CSS, Django (Frontend).
- Django Framework (Backend).
- GitHub (For collaboration).
- VS Code (IDE).

3.2 Hardware Requirements

The following hardware is required for the development of my project:

- Microsoft Windows 7/8/10 (64-bit).
- 4 GB RAM.
- 4 GB disk space.

CHAPTER-IV

METHODOLOGY

4.1 Algorithm or Method Used

4.1.1 Use Case Diagram:

A use case is a software and system engineering term that describes how a user uses a system to accomplish a particular goal. A use case acts as a software modeling technique that defines the features to be implemented and the resolution of any errors that may be encountered. There are three basic elements that make up a use case:

Actors: Actors are the type of users that interact with the system.

System: Use cases capture functional requirements that specify the intended behavior of the system.

Goals: Use cases are typically initiated by a user to fulfill goals, describing the activities and variants involved in attaining the goal.

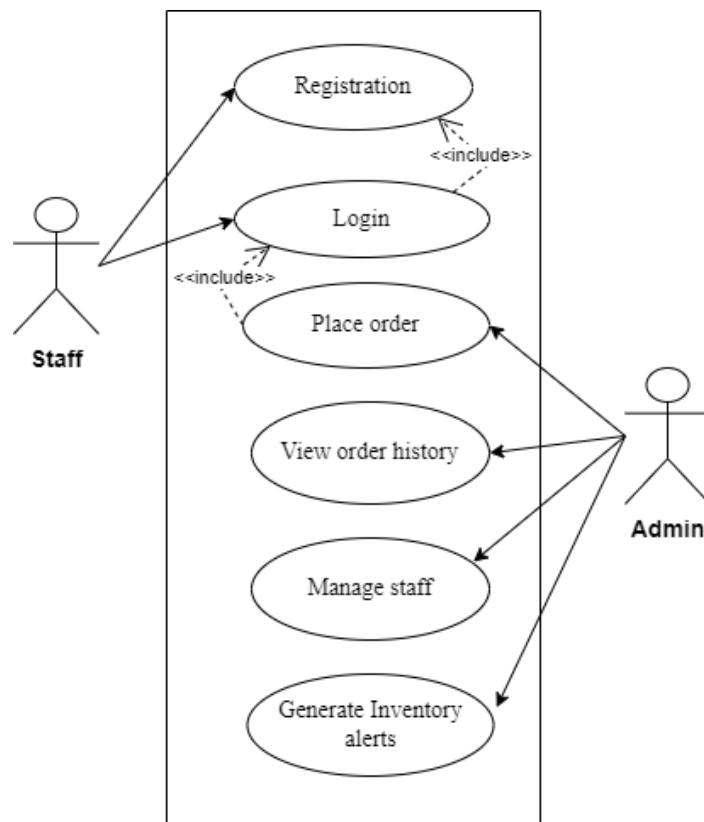


Figure 4.1.1.1 Use case Diagram- Inventory Management

4.1.2 Data Flow Diagram (DFD):

A data flow diagram (DFD) maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles and arrows, plus short text labels, to show data inputs, outputs, storage points and the routes between each destination. Data flowcharts can range from simple, even hand-drawn process overviews, to in-depth, multi-level DFDs that dig progressively deeper into how the data is handled. They can be used to analyze an existing system or model a new one.



Figure 4.1.2. 1 Level zero DFD- Inventory Management

4.1.3 Entity Relationship Diagram

An Entity Relationship (ER) diagram is a type of flowchart that illustrates how “entities” such as people, objects or concepts relate to each other within a system. ER Diagrams are most often used to design or debug relational databases in the fields of software engineering, business information systems, education and research.

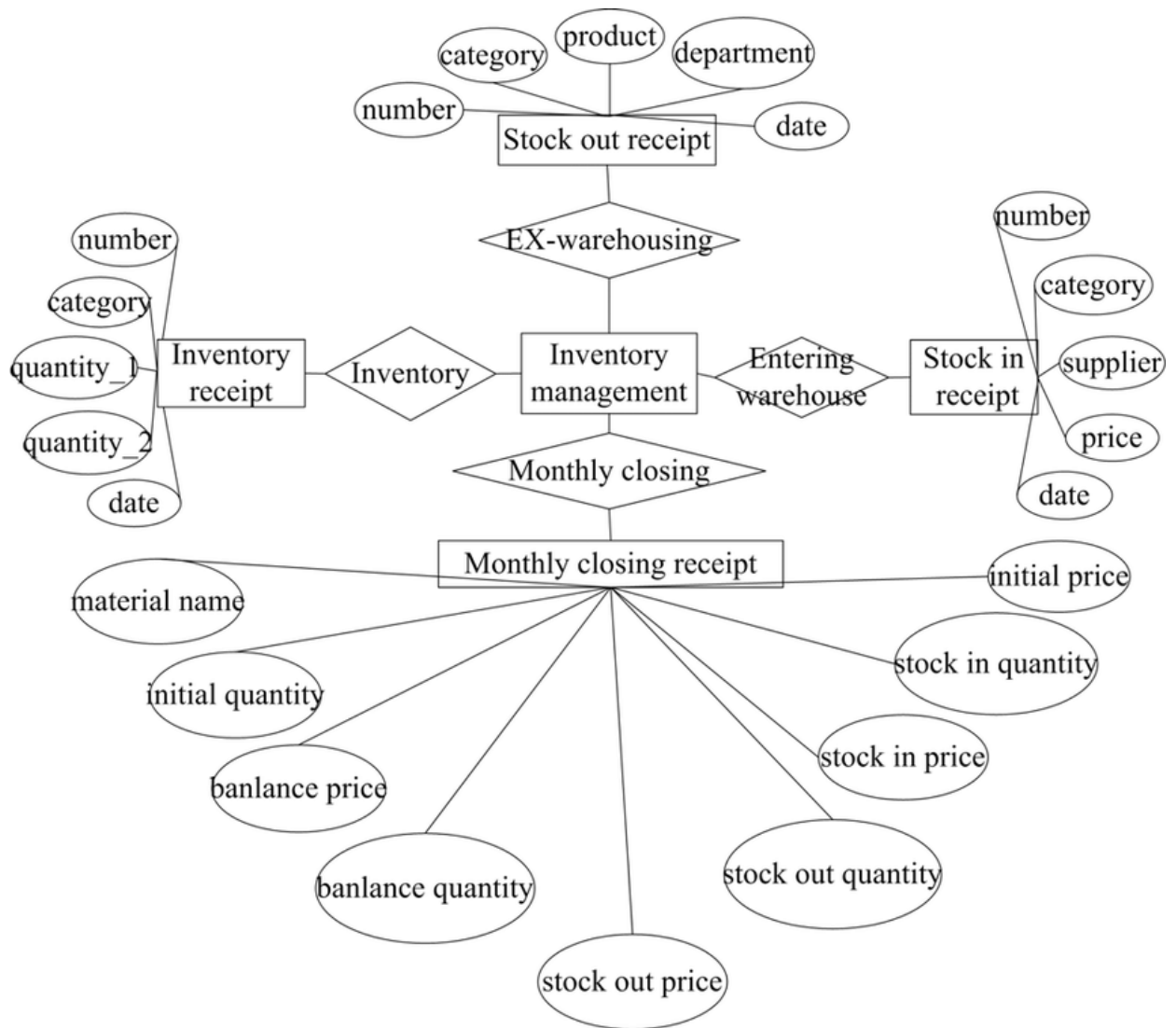


Figure 4.1.3. 1 Inventory Management

4.2 Feasibility Study

The feasibility study is necessary to evaluate the risks and potential benefits of the project in question and decide whether it should go forward or not. Whether it is worthwhile to continue forward with a project depends on its viability. The method utilized to arrive at this result is referred to as a feasibility study. A project's viability and value are determined by this sort of study.

4.2.1 Technical Feasibility:

Determine if the organization's current technological infrastructure is capable of supporting an online inventory management system. Aspects including software compatibility, network connectivity, and device capabilities should be taken into account. Determine whether improvements are required or if the appropriate technologies can be installed.

4.2.2 Economic Feasibility:

Examine the available funding sources for establishing and operating an online inventory management system. Take into account the expenses of purchasing software licenses, upgrading your hardware, getting trained, transferring your data, and continuing maintenance and support. Take a look at these expenses in relation to the system's possible advantages and cost reductions.

4.2.3 Operational Feasibility:

Analyze how effectively the organization's current procedures and workflows mesh with the suggested online inventory management system. Assess if any major modifications or adaptations must be made to fit the new system. Think about the effect on personnel, the need for training, and how it interacts with current business processes.

4.2.4 Scalability:

Evaluate the online inventory control system's capacity for improvement. Evaluate out if it can support the organization's future development and growth goals. Consider about things like how the system will handle expanding inventory numbers, multi-channel sales, and platform integration.

CHAPTER-V

RESULTS

5.1 Results

The major goal of our project is to enable Inventory managements for our users rather than managing data manually. Our System assisted us in accomplishing our goal. The program made it simple to register users, reducing holding cost, minimized the losses and wastages compute and enhances the customer satisfactions. During the development process, I learnt about database connections with API integration to the User Interface. Furthermore, I was able to utilize our theoretical understanding to construct a workable application within the constraints of the project deadline.

CHAPTER-VI

CONCLUSION AND RECCOMENDIATIONS

6.1 Recommendations

The suggested actions for our users include:

1. User Training and Support.
2. Regular Updates and Maintenance.
3. Data Security Measures.

6.2 Conclusion

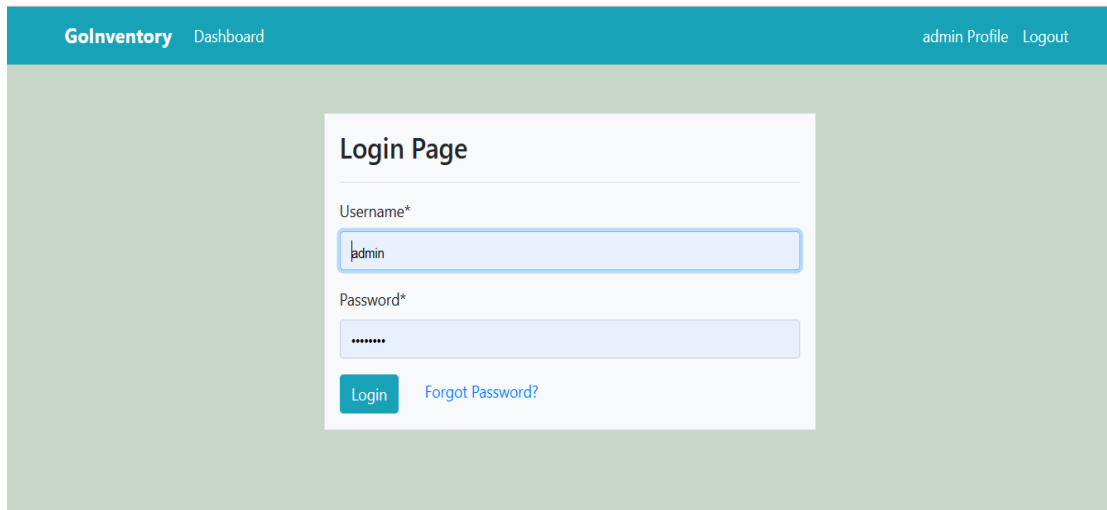
Businesses looking to optimize their inventory procedures and achieve operational excellence have shown to find the creation and implementation of inventory management websites to be essential tools. Real-time tracking, faster operations, increased cost management, and improved decision-making abilities are just a few advantages that our systems provide.

The way firms manage their inventory has changed dramatically as a result of the usage of contemporary technology and user-friendly interfaces in inventory management websites. Websites like these assist organizations to achieve a balance between maintaining ideal inventory levels and satisfying consumer wants, eventually leading to higher customer happiness and profitability. They do this through precise demand forecasts, effective order management, and data-driven insights.

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APPENDIX



The screenshot shows the 'GoInventory' application's login interface. At the top, a teal header bar contains the 'GoInventory' logo, a 'Dashboard' link, and user links for 'admin Profile' and 'Logout'. The main content area has a light green background. Centered on this is a white 'Login Page' box. Inside the box, there are two input fields: 'Username*' with the text 'admin' and 'Password*' with masked characters. Below the password field is a teal 'Login' button and a blue 'Forgot Password?' link.

GoInventory Dashboard admin Profile Logout

Login Page

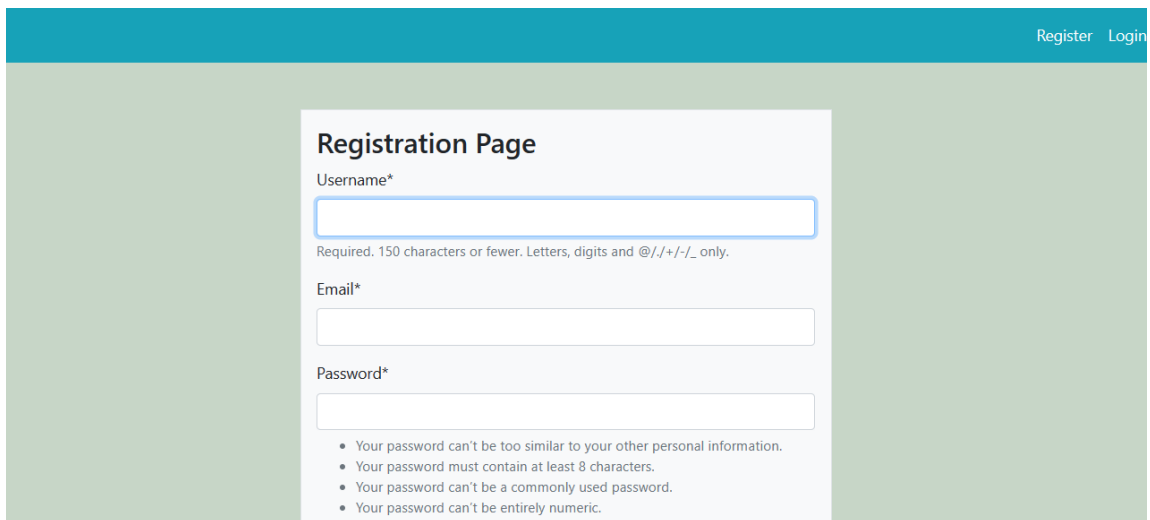
Username*

admin

Password*

Login [Forgot Password?](#)

Figure 6.2. 1 Login Page



The screenshot shows the 'GoInventory' application's registration interface. The teal header bar at the top contains 'Register' and 'Login' links. The main content area has a light green background. Centered on this is a white 'Registration Page' box. It contains three input fields: 'Username*', 'Email*', and 'Password*'. Below the 'Username*' field is a small text requirement: 'Required. 150 characters or fewer. Letters, digits and @/./+/-/_ only.' Below the 'Password*' field is a list of four password requirements.

Register Login

Registration Page

Username*

Required. 150 characters or fewer. Letters, digits and @/./+/-/_ only.

Email*

Password*

- Your password can't be too similar to your other personal information.
- Your password must contain at least 8 characters.
- Your password can't be a commonly used password.
- Your password can't be entirely numeric.

Figure 6.2. 2 Registration Page

Register Login

Password Reset Page

Email*

Reset Password

Figure 6.2. 3 Password Rest Page

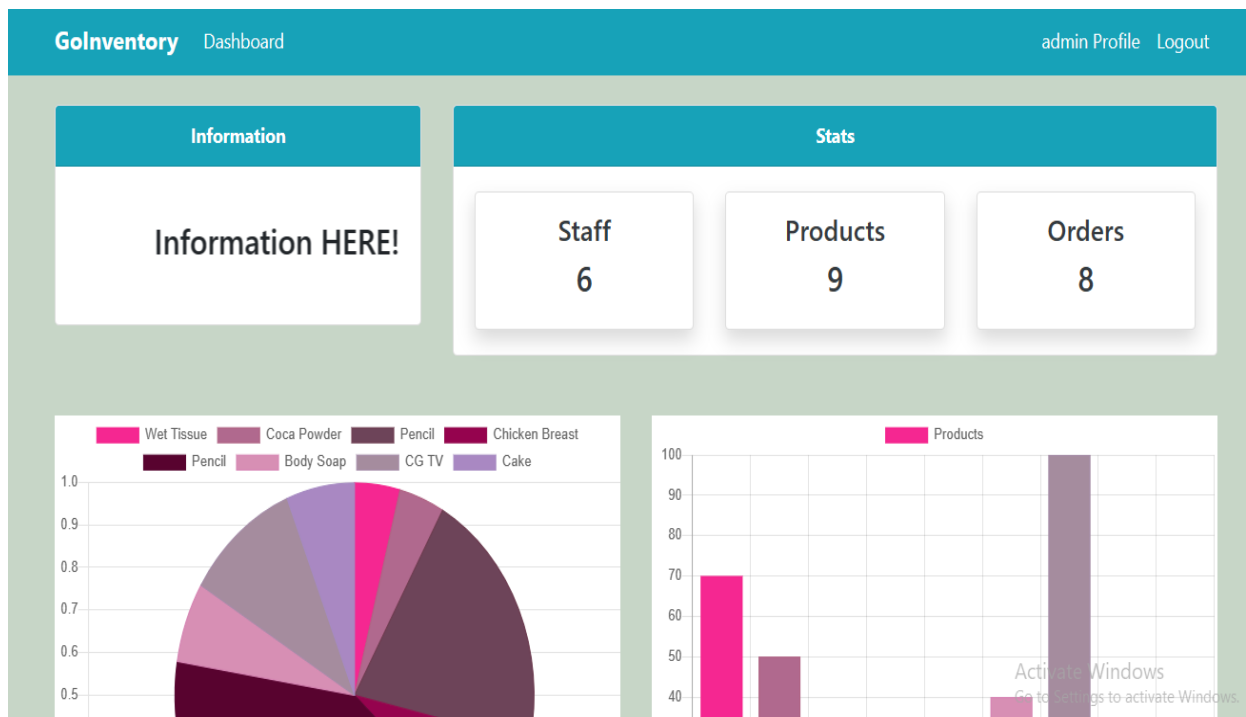


Figure 6.2. 4 Home Page

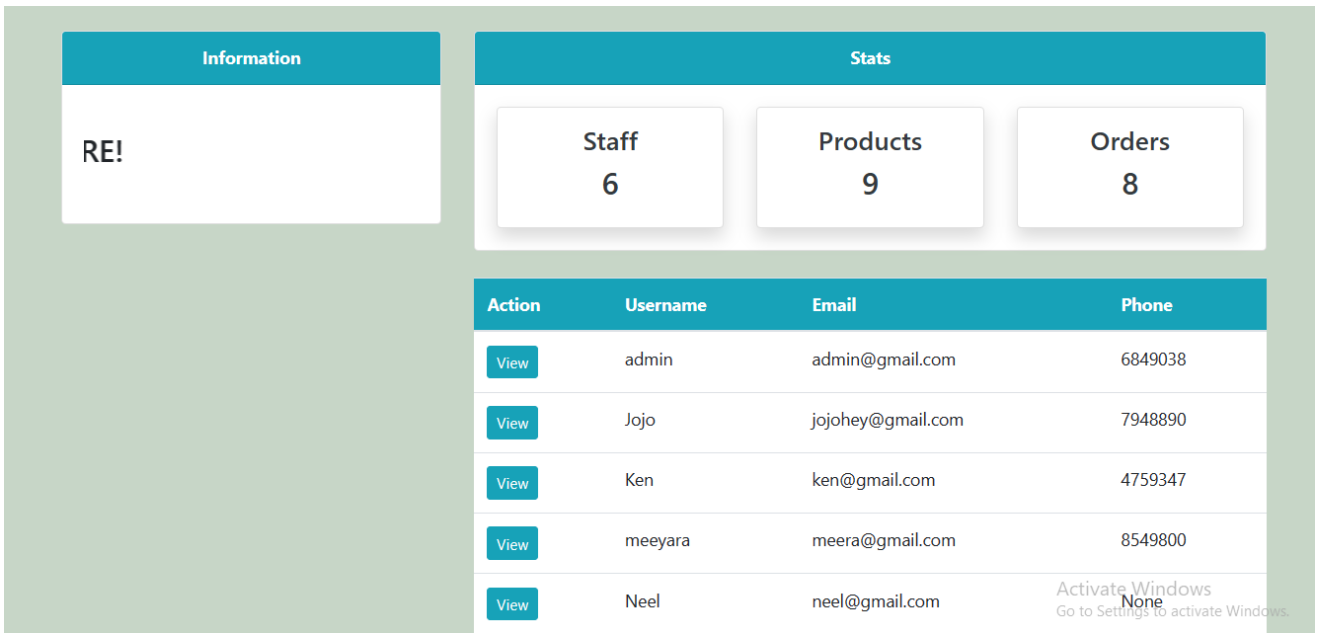


Figure 6.2. 5 Staff Page

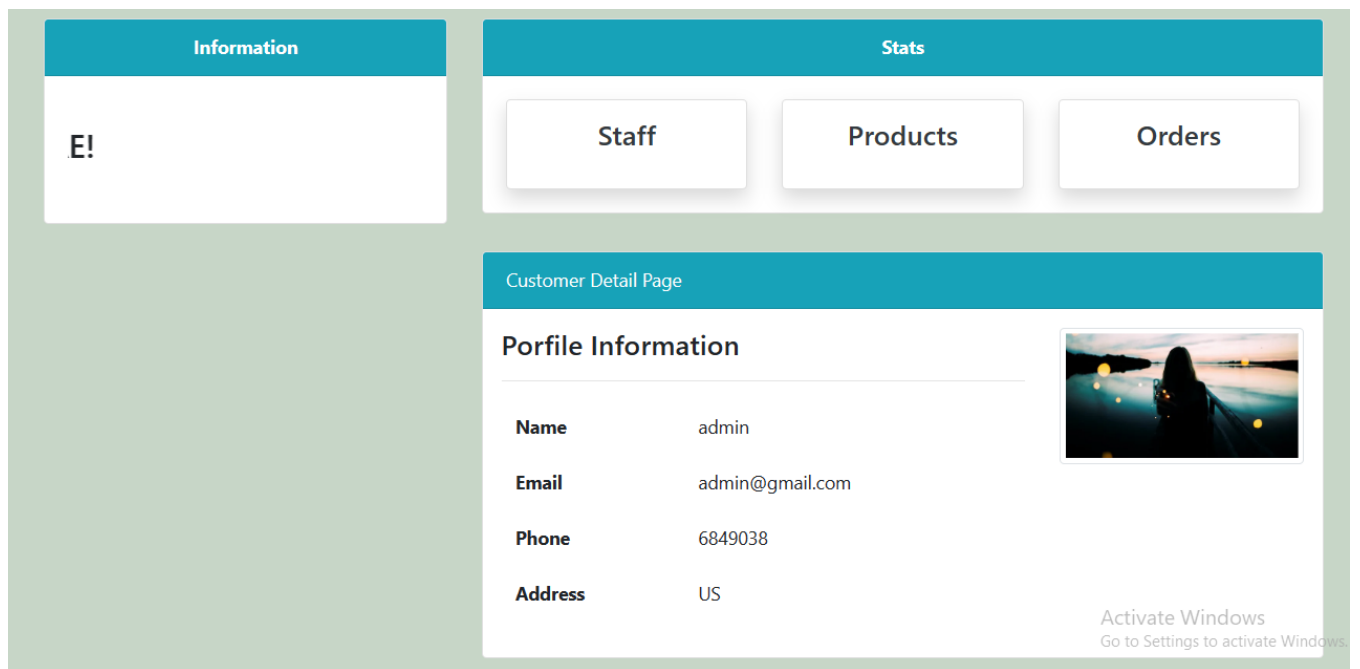


Figure 6.2. 6 Staff View Page

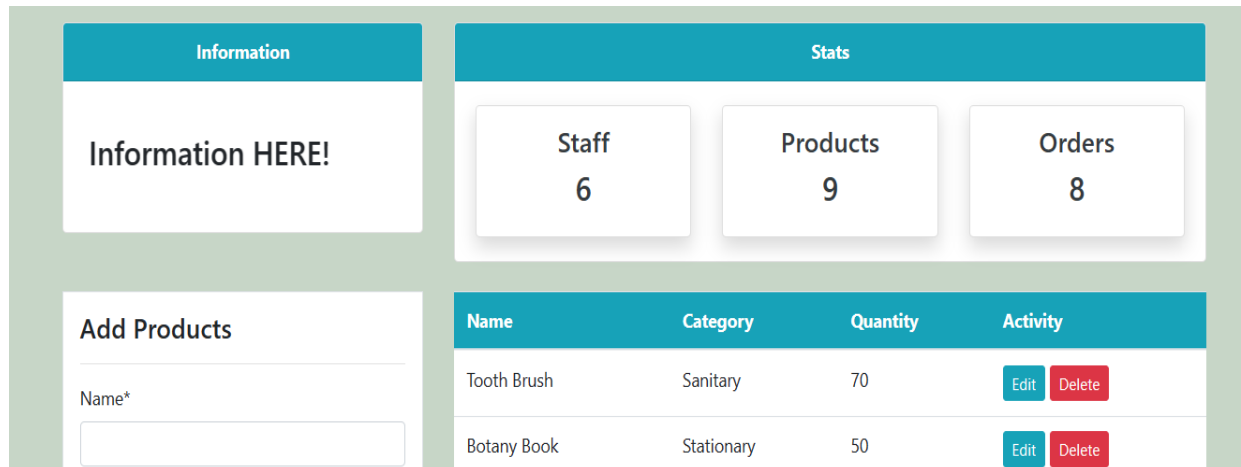


Figure 6.2. 7 Product Page

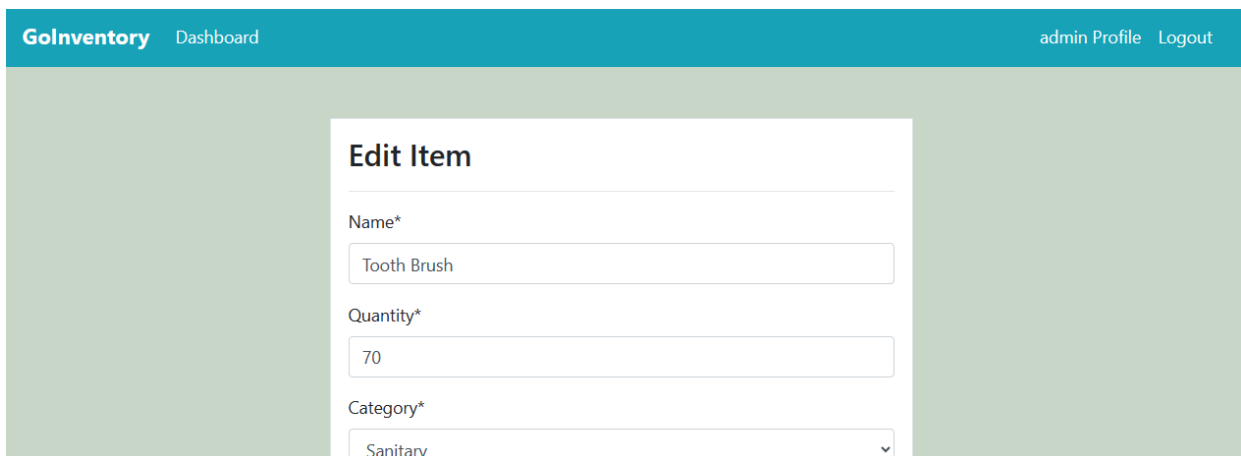


Figure 6.2. 8 Product Edit

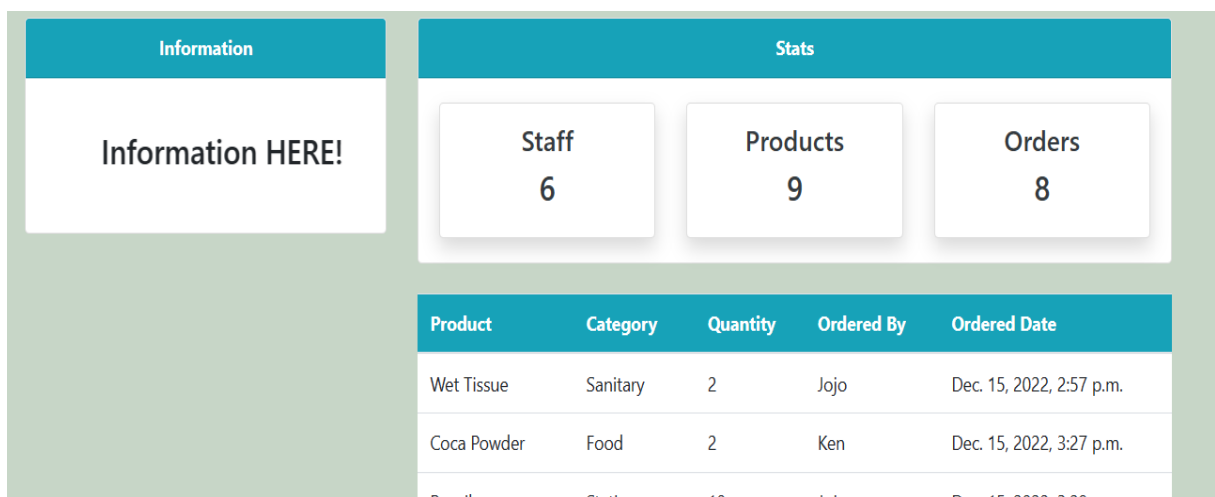


Figure 6.2. 9 Product Order

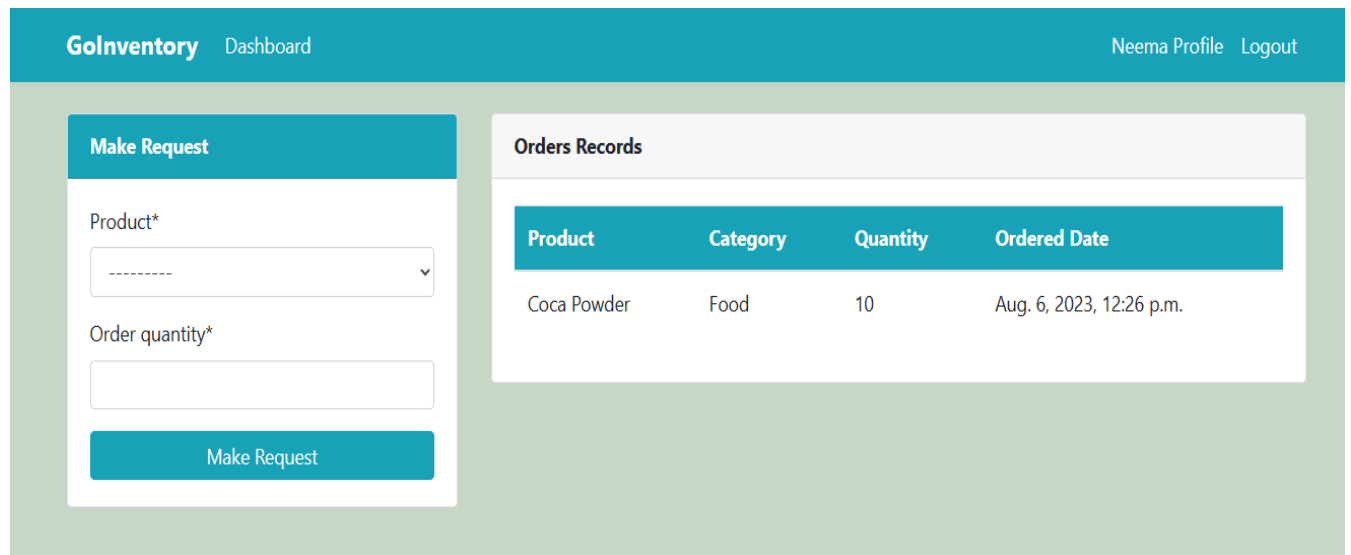


Figure 6.2. 10 User Side Home Page

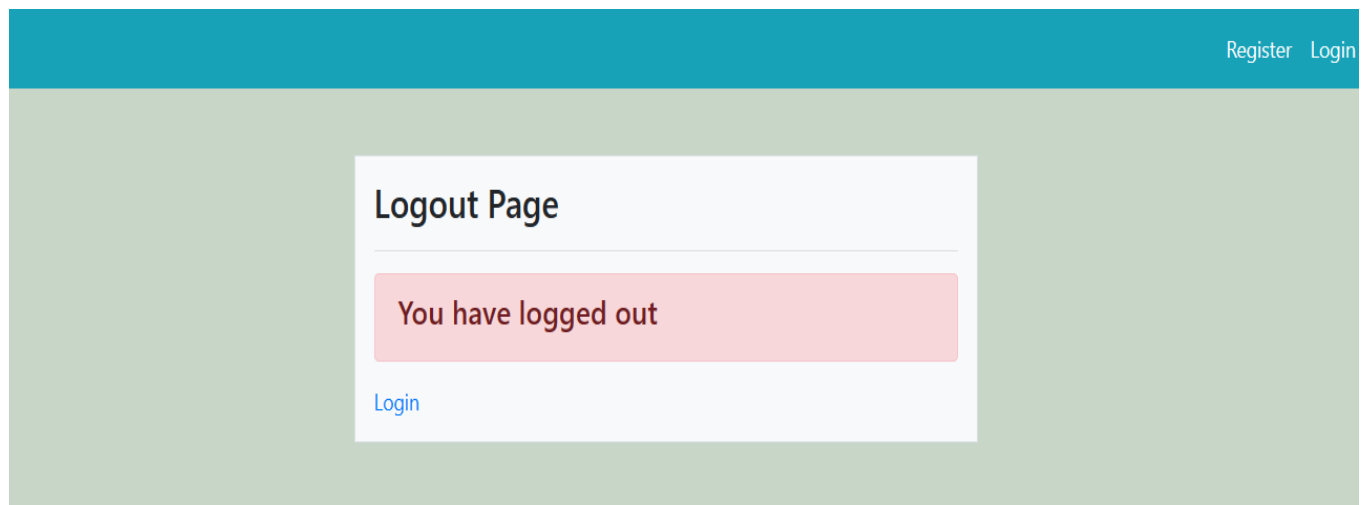


Figure 6.2. 11 Logout Page